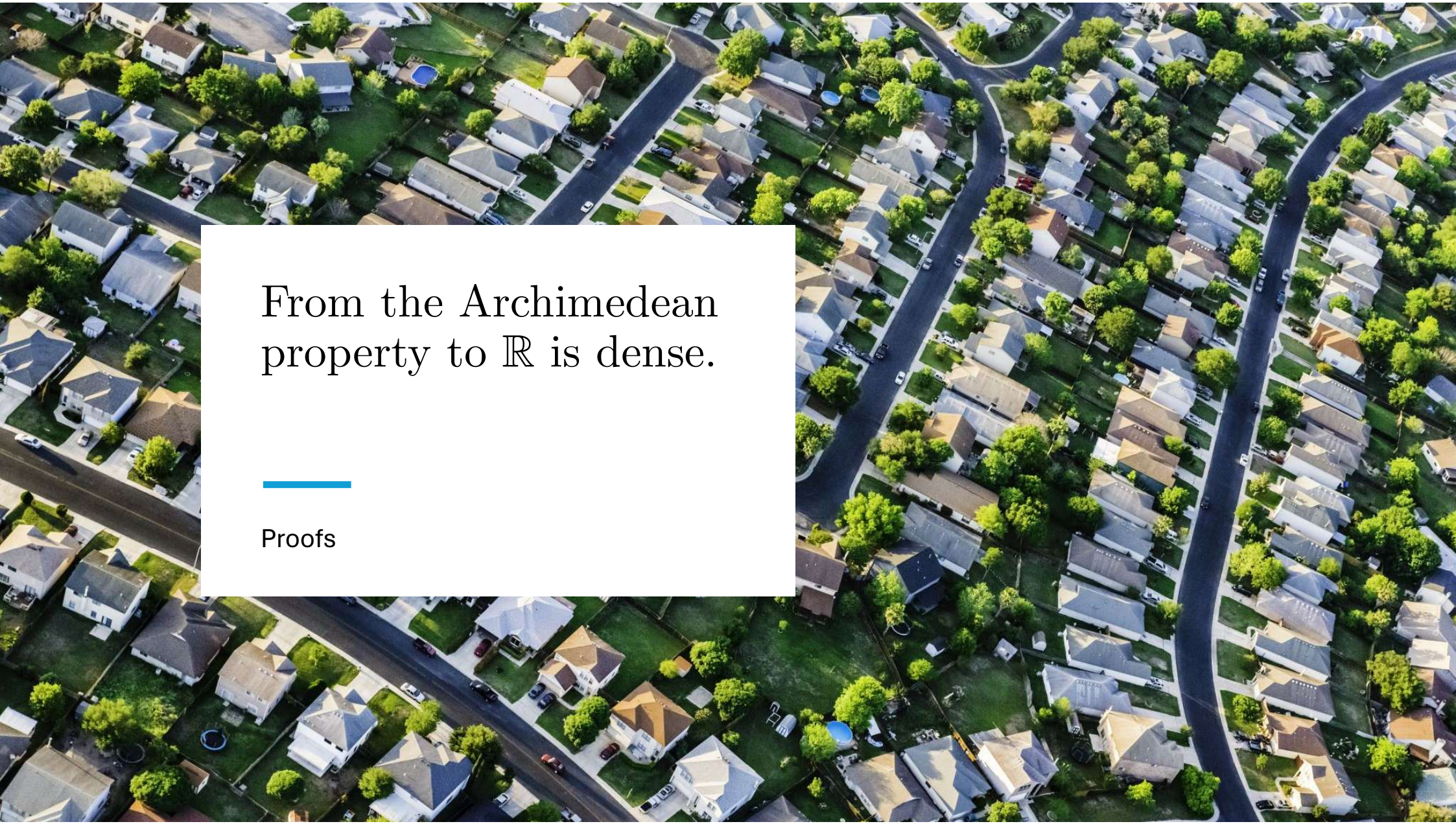


From the Archimedean property to $\mathbb{R}$ is dense.

Proofs

Outline

- Prove the Archimedean Property
- Prove that $\mathbb{Q}$ is dense in $\mathbb{R}$.
- Prove that $\mathbb{R}\backslash\mathbb{Q}$ is dense in $\mathbb{R}$.
- Prove that $\mathbb{R}$ is dense.

Theorem 3.4. Archimedean property:

For any real numbers x and y with $y > x > 0$,

there exists a natural number n such that $y < nx$.

$$\boxed{\forall x, y \in \mathbb{R}^+, y > x, \exists n \in \mathbb{N} \ni y < nx}$$

Proof. Set $z = \frac{y}{x} \in \mathbb{R}^+$. Note that $z > 1$.

$\Rightarrow$ Equivalent to $\forall z \in (1, \infty)\ \exists n \in \mathbb{N} \ni z < n$.

Prove by contradiction, i.e., suppose $\exists z \in (1, \infty) \ni z \geq n\ \forall n \in \mathbb{N}$.

$\Rightarrow$ Set $\mathbb{N}$ is bounded above.

Since $\phi \neq \mathbb{N} \subset \mathbb{R}$, it follows by the completeness axiom that

$\exists \alpha = \sup \mathbb{N} \in \mathbb{R}$.

We show that $\alpha \notin \mathbb{N}$. Suppose $\alpha \in \mathbb{N}$, then

$$\alpha < \alpha + 1 \in \mathbb{N},$$ a contradiction.

This is because $\mathbb{N}$ is bounded above, so it does not contain any element $> \alpha$.

So, $\alpha \in \mathbb{R} \setminus \mathbb{N}$.

Now $\alpha - 1 < \alpha$, and since $\alpha \in \mathbb{R} \setminus \mathbb{N}$, it follows that

$$\exists m \in \mathbb{N} \ni \alpha - 1 < m < \alpha\,.$$ (we elaborate this proof later)

This implies that $\alpha < m + 1 \in \mathbb{N}$, a contradiction since supremum is an upper bound.

Show that $\forall \alpha \in \mathbb{R} \setminus \mathbb{Z}\ \exists m \in \mathbb{Z} \ni \alpha < m < \alpha + 1$.

Proof. Note that the smallest gap between any two integers is one.

Suppose that $\exists \alpha \in \mathbb{R} \setminus \mathbb{Z} \ni m \notin (\alpha, \alpha + 1)\ \forall m \in \mathbb{Z}$.

Now the gap between α and $\alpha + 1$ is one.

It follows that there exists an integer between α and $\alpha + 1$

or otherwise both $\alpha, \alpha + 1 \in \mathbb{Z}$.

Both cases lead to a contradiction.

Corollary 3.5. For any real numbers $\varepsilon > 0$,

there exists a natural number n such that $\varepsilon > \frac{1}{n}$.

$$\boxed{\forall \varepsilon \in \mathbb{R}^+ \; \exists n \in \mathbb{N} \ni \varepsilon > \frac{1}{n}}$$

Theorem 3.7. $\mathbb{Q}$ is dense in $\mathbb{R}$.

$\forall x, y \in \mathbb{R}$ with $x < y$ $\exists z \in \mathbb{Q}$ ∋ $x < z < y$.

Proof. Suppose that $0 \leq x < y$.

Then by the Archimedean property:

$\forall \varepsilon > 0 \; \exists n \in \mathbb{N} \ni \frac{1}{n} < \varepsilon$.

We choose $\varepsilon = y - x > 0$, then $\frac{1}{n} < y - x$, or

$$1 + nx < ny \quad (1)$$

Note that $S \subset \mathbb{N} \subset \mathbb{R}$, $S \neq \phi$ and bounded below. This guarantee the existence of infimum. Here, minimum = infimum.

Let $S = \left\{k \in \mathbb{N} : k > nx\right\}$.

Then $\exists m \in S$ such that m is the smallest such number.

$$m \in S \Rightarrow m > nx \quad (2)$$

$$m - 1 \notin S \Rightarrow m - 1 \leq nx \quad (3)$$

$$\begin{cases} 1 + nx < ny & (1) \\ nx < m & (2) \\ m \leq nx + 1 & (3) \end{cases} \Rightarrow \underset{(2)}{nx} < \underset{(3)}{m \leq} 1 + \underset{(1)}{nx} < ny$$

$$\underset{\div n}{\Rightarrow} x < \frac{m}{n} \leq \frac{1}{n} + x < y$$

Case 2. When $x < 0$ and $y > 0$, choose $z = 0 \in \mathbb{Q}$.

Case 3. When $x < y \leq 0$ it follows that $-x > -y \geq 0$.

By the Case 1, there exists a $z \in \mathbb{Q}$ such that $-y < z < -x$.

Hence, $y > -z > x$.

Recall the result proven in Example 4.5 of lecture notes Chapter 1:

$\sqrt{2}$ is irrational.

Lemma. Prove that $a + r$ is irrational if a is rational and r is irrational.

Prove by contradiction. Let $a + r \in \mathbb{Q}$. Then $a + r = \frac{c}{d}$ for some $c, d \in \mathbb{Z}$, $d \neq 0$.

Since $a = \frac{m}{n}, m, n \in \mathbb{Z}, n \neq 0$, it follows that $r = \frac{c}{d} - \frac{m}{n} = \frac{cn-md}{nd} \in \mathbb{Q}$, a contradiction.

Corollary 3.8. $\mathbb{R} \setminus \mathbb{Q}$ is dense in $\mathbb{R}$.

$\forall a, b \in \mathbb{R}$ with $a < b$, $\exists c \in \mathbb{R} \setminus \mathbb{Q} \ni a < c < b$.

Proof. Let $a, b \in \mathbb{R}$ with $a < b$.

It follows that $a - \sqrt{2} < b - \sqrt{2}$

and note that both $a - \sqrt{2}, b - \sqrt{2} \in \mathbb{R}$.

By the density of $\mathbb{Q}$ in $\mathbb{R}$ (Theorem 3.7),

there exists $c \in \mathbb{Q}$ such that

$a - \sqrt{2} < c < b - \sqrt{2} \Rightarrow a < c + \sqrt{2} < b$.

Since $c + \sqrt{2} \notin \mathbb{Q}$, it follows that

irrational numbers are dense in the real numbers.

Combining Theorem 3.7 and Corollary 3.8, we have $\mathbb{R}$ is dense.